SCIENCE

Basic 6

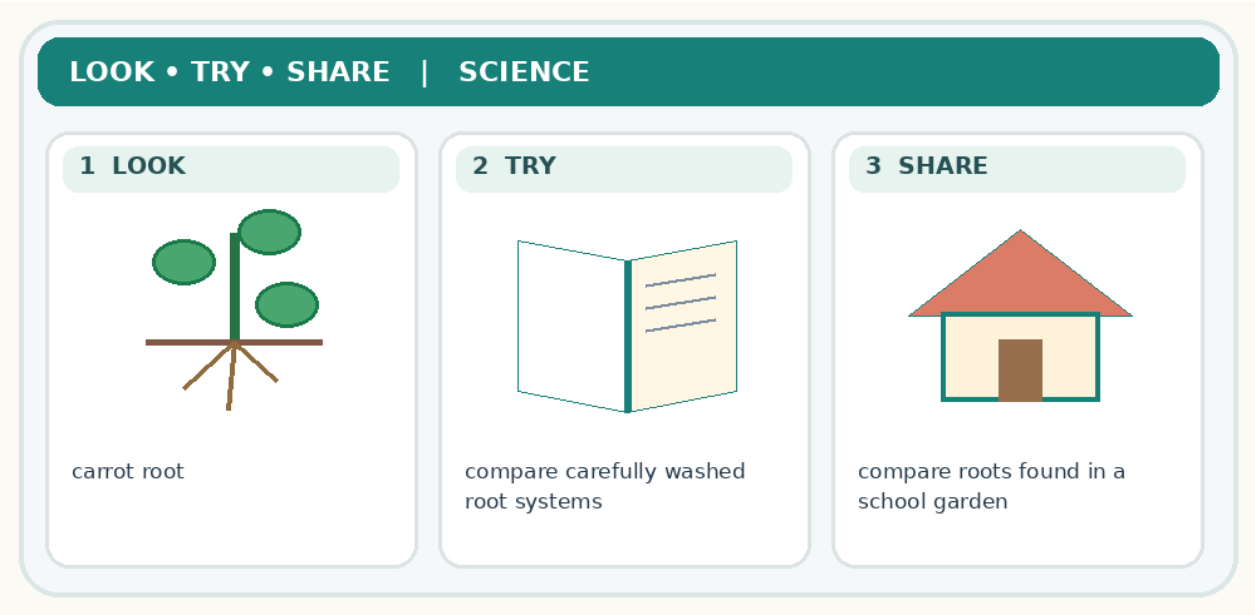
14 illustrated learning units • learner activities • 5 Play + 10 Challenge questions per lesson

Ghana-centred multimedia companion; independently authored and aligned at topic/theme level. Not an official NaCCA publication.

Open Offline_Interactive.html after extracting the package to play the matching narrated MP4s inside the lesson pages.

Lesson 1. Tap roots and fibrous roots

Diversity of Matter | Term 1



EXPLORE

Tap roots have one main root; fibrous roots have many thin roots.

Look for: carrot root, grass root, bean root, maize root

PLAY & APPLY

Activity: compare carefully washed root systems

Materials: uprooted weeds and drawing paper

Real life: compare roots found in a school garden

Remember: Tap roots and fibrous roots are not identical shapes.

Partner talk: What did you notice? Can you give a reason for your answer?

PLAY QUESTIONS

1. Which example is connected with this lesson about Tap roots and fibrous roots?

A. shade B. carrot root C. wind D. handwashing

2. What is the best activity for practising Tap roots and fibrous roots?

A. ignore the instructions and rush B. compare carefully washed root systems C. leave all materials untouched D. copy without discussing anything

3. Which other example helps us explore Tap roots and fibrous roots?

A. frog B. grass root C. clean wound D. plants

4. Which materials should the group prepare to study Tap roots and fibrous roots?

A. uprooted weeds and drawing paper B. nothing suitable for the task C. unsafe tools without an adult D. broken objects found at random

5. Where can you use what you learnt about Tap roots and fibrous roots?

A. avoid talking about the idea B. compare roots found in a school garden C. ignore the safety guidance D. damage the classroom display

CHALLENGE QUESTIONS

1. Which explanation best describes our main idea in Tap roots and fibrous roots?

A. An observation is exactly the same as a guess. B. Tap roots have one main root; fibrous roots have many thin roots. C. One example proves that all cases are identical. D. Safety instructions can be ignored during an activity.

2. A classmate makes a mistake in Tap roots and fibrous roots. Which reminder helps?

A. An observation is exactly the same as a guess. B. Safety instructions can be ignored during an activity. C. One example proves that all cases are identical. D. Tap roots and fibrous roots are not identical shapes.

3. The class plans to explore Tap roots and fibrous roots. Which action fits?

A. compare carefully washed root systems B. mix unrelated ideas without planning C. skip the task and claim it is complete D. guess without looking at evidence

4. Choose a pair of examples that fits Tap roots and fibrous roots.

A. shade and stones B. carrot root and maize root C. wind and handwashing D. steel and lever

5. Which materials are most helpful for Tap roots and fibrous roots?

A. unrelated broken electronics B. random objects that cannot help the task C. uprooted weeds and drawing paper D. sharp tools without supervision

6. How could a learner apply Tap roots and fibrous roots outside school?

A. refuse to practise the lesson B. leave a mess after the lesson C. ignore school or home safety rules D. compare roots found in a school garden

7. Which key word should appear on a Tap roots and fibrous roots revision card?

A. clay B. snake C. bean root D. Earth

8. While studying Tap roots and fibrous roots, which choice shows careful learning?

A. observe, try, explain and check your result B. copy answers without understanding C. change the answer without reviewing D. rush without listening

9. Which label belongs on a poster about Tap roots and fibrous roots?

A. maize root B. pollination C. pulley D. trees

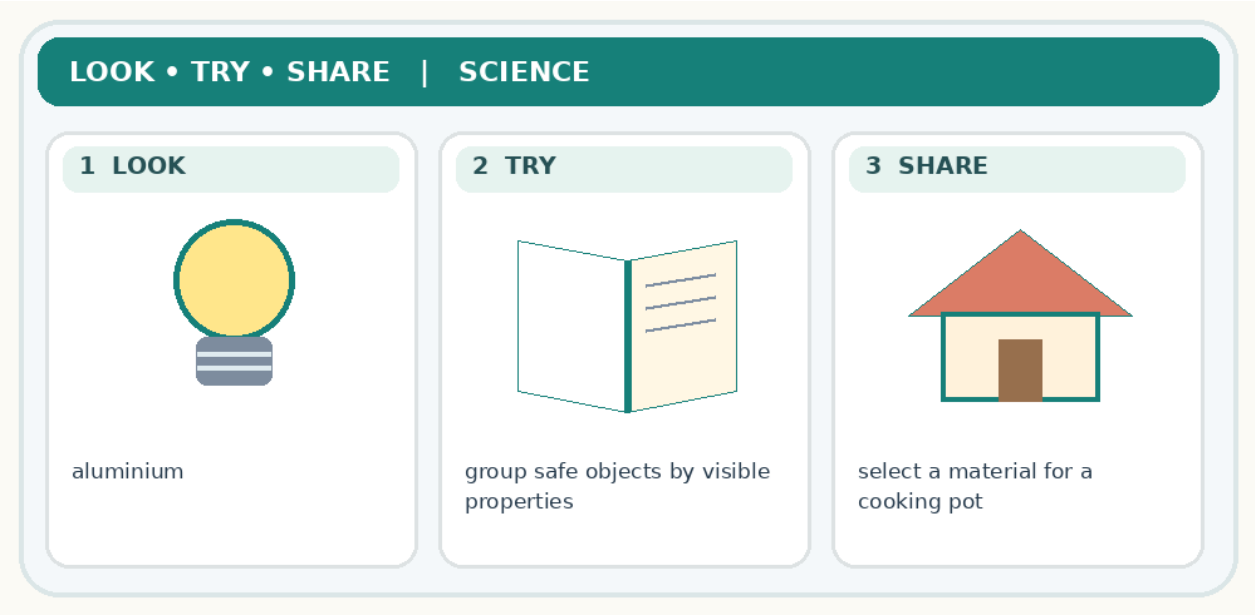
10. After practising Tap roots and fibrous roots, what is a useful next step?

A. claim that feedback is never helpful B. tear up other learners work C. leave the group without sharing D. share one observation and listen to feedback

Discuss the reasons for your choices with a partner. Practical performance and teacher observation matter as well as quiz answers.

Lesson 2. Properties of metals

Diversity of Matter | Term 1



EXPLORE

Many metals are shiny, can be shaped and conduct heat or electricity.

Look for: aluminium, iron, copper, steel

PLAY & APPLY

Activity: group safe objects by visible properties

Materials: safe metal objects and paper samples

Real life: select a material for a cooking pot

Remember: Not all shiny objects are metals.

Partner talk: What did you notice? Can you give a reason for your answer?

PLAY QUESTIONS

1. Which example is connected with this lesson about Properties of metals?

A. clean wound B. seed C. aluminium D. clean drains

2. What is the best activity for practising Properties of metals?

A. copy without discussing anything B. group safe objects by visible properties C. leave all materials untouched D. ignore the instructions and rush

3. Which other example helps us explore Properties of metals?

A. stones B. muscles C. iron D. bean root

4. Which materials should the group prepare to study Properties of metals?

A. unsafe tools without an adult B. safe metal objects and paper samples C. nothing suitable for the task D. broken objects found at random

5. Where can you use what you learnt about Properties of metals?

A. select a material for a cooking pot B. avoid talking about the idea C. ignore the safety guidance D. damage the classroom display

CHALLENGE QUESTIONS

1. Which explanation best describes our main idea in Properties of metals?

A. Many metals are shiny, can be shaped and conduct heat or electricity. B. An observation is exactly the same as a guess. C. One example proves that all cases are identical. D. Safety instructions can be ignored during an activity.

2. A classmate makes a mistake in Properties of metals. Which reminder helps?

A. An observation is exactly the same as a guess. B. One example proves that all cases are identical. C. Not all shiny objects are metals. D. Safety instructions can be ignored during an activity.

3. The class plans to explore Properties of metals. Which action fits?

A. guess without looking at evidence B. skip the task and claim it is complete C. group safe objects by visible properties D. mix unrelated ideas without planning

4. Choose a pair of examples that fits Properties of metals.

A. clean wound and seed B. aluminium and steel C. clean drains and rest D. snake and plants

5. Which materials are most helpful for Properties of metals?

A. safe metal objects and paper samples B. unrelated broken electronics C. sharp tools without supervision D. random objects that cannot help the task

6. How could a learner apply Properties of metals outside school?

A. select a material for a cooking pot B. refuse to practise the lesson C. leave a mess after the lesson D. ignore school or home safety rules

7. Which key word should appear on a Properties of metals revision card?

A. Moon B. pollination C. maize root D. copper

8. While studying Properties of metals, which choice shows careful learning?

A. observe, try, explain and check your result B. rush without listening C. change the answer without reviewing D. copy answers without understanding

9. Which label belongs on a poster about Properties of metals?

A. irrigation B. steel C. loam D. clean water

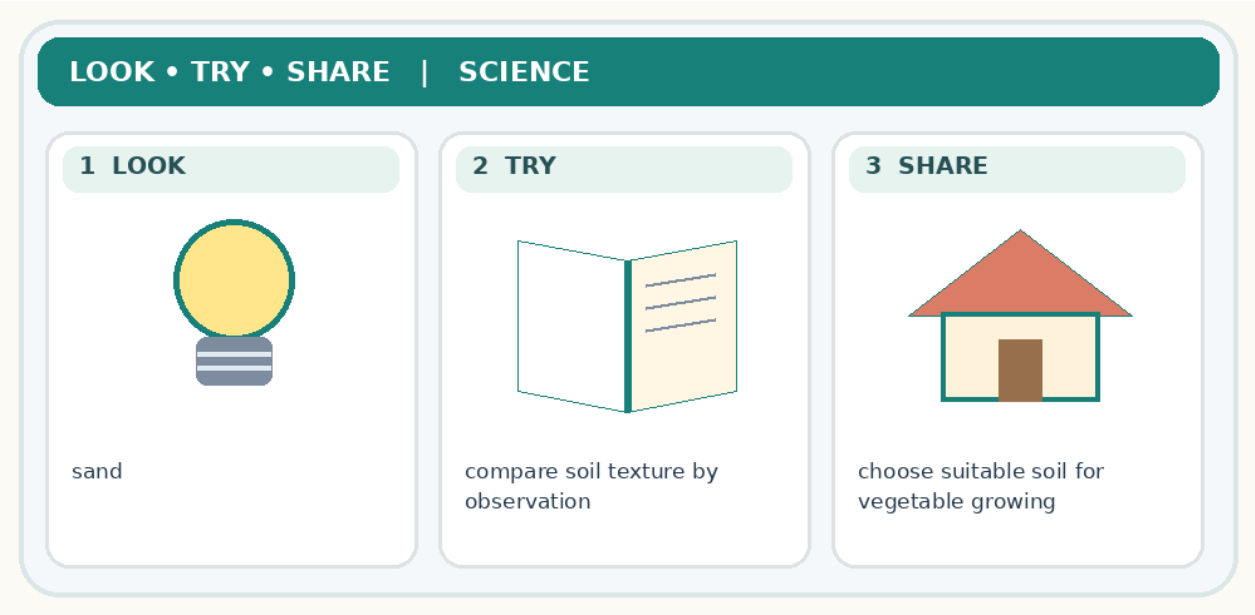
10. After practising Properties of metals, what is a useful next step?

A. share one observation and listen to feedback B. claim that feedback is never helpful C. leave the group without sharing D. tear up other learners work

Discuss the reasons for your choices with a partner. Practical performance and teacher observation matter as well as quiz answers.

Lesson 3. Soil and rocks

Cycles | Term 1



EXPLORE

Soils contain particles and organic matter and differ in water holding.

Look for: sand, clay, loam, stones

PLAY & APPLY

Activity: compare soil texture by observation

Materials: soil samples and transparent cups

Real life: choose suitable soil for vegetable growing

Remember: Sand and clay do not hold water equally.

Partner talk: What did you notice? Can you give a reason for your answer?

PLAY QUESTIONS

1. Which example is connected with this lesson about Soil and rocks?

A. wind B. pulley C. sun D. sand

2. What is the best activity for practising Soil and rocks?

A. leave all materials untouched B. copy without discussing anything C. compare soil texture by observation D. ignore the instructions and rush

3. Which other example helps us explore Soil and rocks?

A. mosquito net B. grass root C. clay D. maize root

4. Which materials should the group prepare to study Soil and rocks?

A. nothing suitable for the task B. unsafe tools without an adult C. broken objects found at random D. soil samples and transparent cups

5. Where can you use what you learnt about Soil and rocks?

A. ignore the safety guidance B. choose suitable soil for vegetable growing C. avoid talking about the idea D. damage the classroom display

CHALLENGE QUESTIONS

1. Which explanation best describes our main idea in Soil and rocks?

A. Soils contain particles and organic matter and differ in water holding. B. An observation is exactly the same as a guess. C. One example proves that all cases are identical. D. Safety instructions can be ignored during an activity.

2. A classmate makes a mistake in Soil and rocks. Which reminder helps?

A. One example proves that all cases are identical. B. Sand and clay do not hold water equally. C. An observation is exactly the same as a guess. D. Safety instructions can be ignored during an activity.

3. The class plans to explore Soil and rocks. Which action fits?

A. compare soil texture by observation B. skip the task and claim it is complete C. guess without looking at evidence D. mix unrelated ideas without planning

4. Choose a pair of examples that fits Soil and rocks.

A. pulley and sun B. wind and trees C. handwashing and carrot root D. sand and stones

5. Which materials are most helpful for Soil and rocks?

A. sharp tools without supervision B. random objects that cannot help the task C. soil samples and transparent cups D. unrelated broken electronics

6. How could a learner apply Soil and rocks outside school?

A. ignore school or home safety rules B. refuse to practise the lesson C. choose suitable soil for vegetable growing D. leave a mess after the lesson

7. Which key word should appear on a Soil and rocks revision card?

A. irrigation B. flower C. loam D. water filter

8. While studying Soil and rocks, which choice shows careful learning?

A. copy answers without understanding B. rush without listening C. observe, try, explain and check your result D. change the answer without reviewing

9. Which label belongs on a poster about Soil and rocks?

A. stones B. iron C. plants D. aluminium

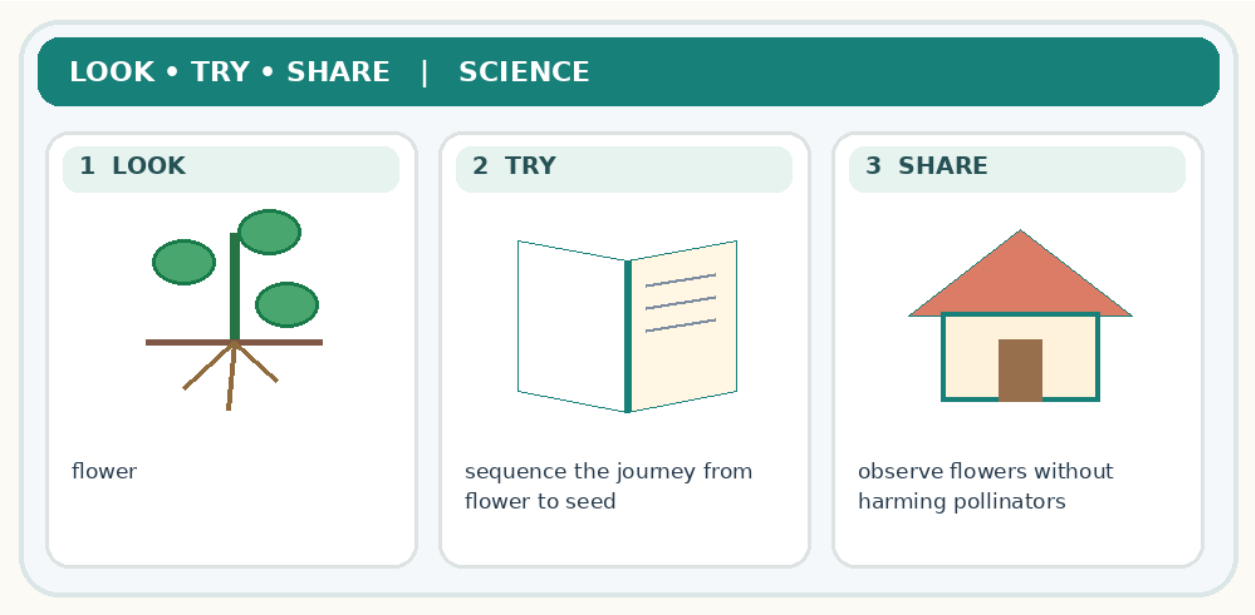
10. After practising Soil and rocks, what is a useful next step?

A. leave the group without sharing B. share one observation and listen to feedback C. tear up other learners work D. claim that feedback is never helpful

Discuss the reasons for your choices with a partner. Practical performance and teacher observation matter as well as quiz answers.

Lesson 4. Plant life cycles and reproduction

Cycles | Term 1



EXPLORE

Flowering plants make seeds that can form new plants.

Look for: flower, pollination, seed, fruit

PLAY & APPLY

Activity: sequence the journey from flower to seed

Materials: flower pictures and life-cycle cards

Real life: observe flowers without harming pollinators

Remember: A fruit may contain seeds; it is not the same as a root.

Partner talk: What did you notice? Can you give a reason for your answer?

PLAY QUESTIONS

1. Which example is connected with this lesson about Plant life cycles and reproduction?

A. coal B. bean root C. plants D. flower

2. What is the best activity for practising Plant life cycles and reproduction?

A. copy without discussing anything B. sequence the journey from flower to seed C. ignore the instructions and rush D. leave all materials untouched

3. Which other example helps us explore Plant life cycles and reproduction?

A. pollination B. snake C. lungs D. vaccination

4. Which materials should the group prepare to study Plant life cycles and reproduction?

A. unsafe tools without an adult B. broken objects found at random C. flower pictures and life-cycle cards D. nothing suitable for the task

5. Where can you use what you learnt about Plant life cycles and reproduction?

A. ignore the safety guidance B. observe flowers without harming pollinators C. avoid talking about the idea D. damage the classroom display

CHALLENGE QUESTIONS

1. Which explanation best describes our main idea in Plant life cycles and reproduction?

A. Safety instructions can be ignored during an activity. B. One example proves that all cases are identical. C. An observation is exactly the same as a guess. D. Flowering plants make seeds that can form new plants.

2. A classmate makes a mistake in Plant life cycles and reproduction. Which reminder helps?

A. An observation is exactly the same as a guess. B. Safety instructions can be ignored during an activity. C. A fruit may contain seeds; it is not the same as a root. D. One example proves that all cases are identical.

3. The class plans to explore Plant life cycles and reproduction. Which action fits?

A. mix unrelated ideas without planning B. guess without looking at evidence C. skip the task and claim it is complete D. sequence the journey from flower to seed

4. Choose a pair of examples that fits Plant life cycles and reproduction.

A. bean root and plants B. coal and wheel C. flower and fruit D. lever and loam

5. Which materials are most helpful for Plant life cycles and reproduction?

A. unrelated broken electronics B. sharp tools without supervision C. flower pictures and life-cycle cards D. random objects that cannot help the task

6. How could a learner apply Plant life cycles and reproduction outside school?

A. observe flowers without harming pollinators B. leave a mess after the lesson C. refuse to practise the lesson D. ignore school or home safety rules

7. Which key word should appear on a Plant life cycles and reproduction revision card?

A. mosquito net B. iron C. seed D. clean wound

8. While studying Plant life cycles and reproduction, which choice shows careful learning?

A. change the answer without reviewing B. rush without listening C. observe, try, explain and check your result D. copy answers without understanding

9. Which label belongs on a poster about Plant life cycles and reproduction?

A. fruit B. sun C. maize root D. bandage

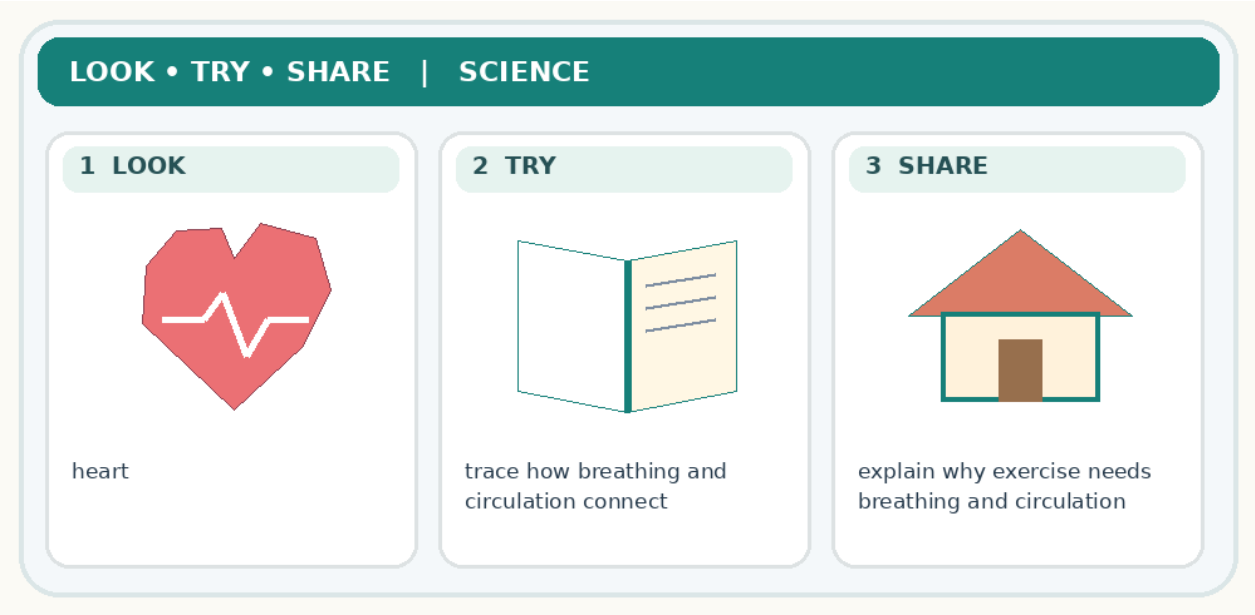
10. After practising Plant life cycles and reproduction, what is a useful next step?

A. share one observation and listen to feedback B. tear up other learners work C. claim that feedback is never helpful D. leave the group without sharing

Discuss the reasons for your choices with a partner. Practical performance and teacher observation matter as well as quiz answers.

Lesson 5. Body systems working together

Systems | Term 1



EXPLORE

Organs and body systems work together to support life.

Look for: heart, lungs, muscles, bones

PLAY & APPLY

Activity: trace how breathing and circulation connect

Materials: body outline and labels

Real life: explain why exercise needs breathing and circulation

Remember: No single organ can do every job alone.

Partner talk: What did you notice? Can you give a reason for your answer?

PLAY QUESTIONS

1. Which example is connected with this lesson about Body systems working together?

A. grasshopper B. heart C. Earth D. rain collection

2. What is the best activity for practising Body systems working together?

A. leave all materials untouched B. trace how breathing and circulation connect C. copy without discussing anything D. ignore the instructions and rush

3. Which other example helps us explore Body systems working together?

A. irrigation B. lungs C. coal D. rest

4. Which materials should the group prepare to study Body systems working together?

A. body outline and labels B. unsafe tools without an adult C. nothing suitable for the task D. broken objects found at random

5. Where can you use what you learnt about Body systems working together?

A. ignore the safety guidance B. avoid talking about the idea C. damage the classroom display D. explain why exercise needs breathing and circulation

CHALLENGE QUESTIONS

1. Which explanation best describes our main idea in Body systems working together?

A. Organs and body systems work together to support life. B. Safety instructions can be ignored during an activity. C. An observation is exactly the same as a guess. D. One example proves that all cases are identical.

2. A classmate makes a mistake in Body systems working together. Which reminder helps?

A. One example proves that all cases are identical. B. An observation is exactly the same as a guess. C. No single organ can do every job alone. D. Safety instructions can be ignored during an activity.

3. The class plans to explore Body systems working together. Which action fits?

A. skip the task and claim it is complete B. trace how breathing and circulation connect C. guess without looking at evidence D. mix unrelated ideas without planning

4. Choose a pair of examples that fits Body systems working together.

A. heart and bones B. bandage and maize root C. grasshopper and Earth D. rain collection and metal

5. Which materials are most helpful for Body systems working together?

A. body outline and labels B. sharp tools without supervision C. random objects that cannot help the task D. unrelated broken electronics

6. How could a learner apply Body systems working together outside school?

A. ignore school or home safety rules B. leave a mess after the lesson C. refuse to practise the lesson D. explain why exercise needs breathing and circulation

7. Which key word should appear on a Body systems working together revision card?

A. muscles B. loam C. aluminium D. sand

8. While studying Body systems working together, which choice shows careful learning?

A. change the answer without reviewing B. copy answers without understanding C. rush without listening D. observe, try, explain and check your result

9. Which label belongs on a poster about Body systems working together?

A. bones B. solar dryer C. snake D. handwashing

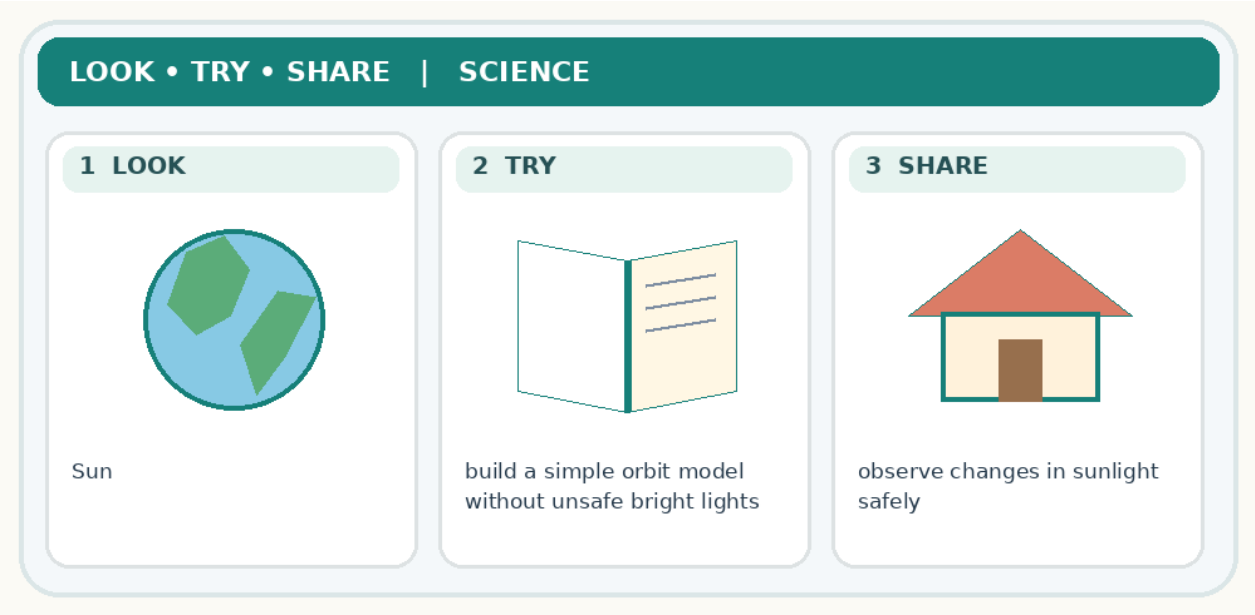
10. After practising Body systems working together, what is a useful next step?

A. leave the group without sharing B. tear up other learners work C. claim that feedback is never helpful D. share one observation and listen to feedback

Discuss the reasons for your choices with a partner. Practical performance and teacher observation matter as well as quiz answers.

Lesson 6. Earth in the solar system

Systems | Term 2



EXPLORE

Earth is one planet orbiting the Sun in the solar system.

Look for: Sun, Earth, Mars, Moon

PLAY & APPLY

Activity: build a simple orbit model without unsafe bright lights

Materials: planet picture cards and string circles

Real life: observe changes in sunlight safely

Remember: The Moon is not a planet.

Partner talk: What did you notice? Can you give a reason for your answer?

PLAY QUESTIONS

1. Which example is connected with this lesson about Earth in the solar system?

A. frog B. clay C. inclined plane D. Sun

2. What is the best activity for practising Earth in the solar system?

A. copy without discussing anything B. build a simple orbit model without unsafe bright lights C. ignore the instructions and rush D. leave all materials untouched

3. Which other example helps us explore Earth in the solar system?

A. maize root B. water filter C. petroleum D. Earth

4. Which materials should the group prepare to study Earth in the solar system?

A. broken objects found at random B. unsafe tools without an adult C. nothing suitable for the task D. planet picture cards and string circles

5. Where can you use what you learnt about Earth in the solar system?

A. observe changes in sunlight safely B. avoid talking about the idea C. damage the classroom display D. ignore the safety guidance

CHALLENGE QUESTIONS

1. Which explanation best describes our main idea in Earth in the solar system?

A. An observation is exactly the same as a guess. B. One example proves that all cases are identical. C. Safety instructions can be ignored during an activity. D. Earth is one planet orbiting the Sun in the solar system.

2. A classmate makes a mistake in Earth in the solar system. Which reminder helps?

A. Safety instructions can be ignored during an activity. B. The Moon is not a planet. C. One example proves that all cases are identical. D. An observation is exactly the same as a guess.

3. The class plans to explore Earth in the solar system. Which action fits?

A. mix unrelated ideas without planning B. guess without looking at evidence C. build a simple orbit model without unsafe bright lights D. skip the task and claim it is complete

4. Choose a pair of examples that fits Earth in the solar system.

A. inclined plane and flower B. clay and frog C. loam and seed D. Sun and Moon

5. Which materials are most helpful for Earth in the solar system?

A. unrelated broken electronics B. sharp tools without supervision C. planet picture cards and string circles D. random objects that cannot help the task

6. How could a learner apply Earth in the solar system outside school?

A. leave a mess after the lesson B. observe changes in sunlight safely C. refuse to practise the lesson D. ignore school or home safety rules

7. Which key word should appear on a Earth in the solar system revision card?

A. plastic B. trees C. fruit D. Mars

8. While studying Earth in the solar system, which choice shows careful learning?

A. change the answer without reviewing B. rush without listening C. copy answers without understanding D. observe, try, explain and check your result

9. Which label belongs on a poster about Earth in the solar system?

A. Moon B. vaccination C. plants D. iron

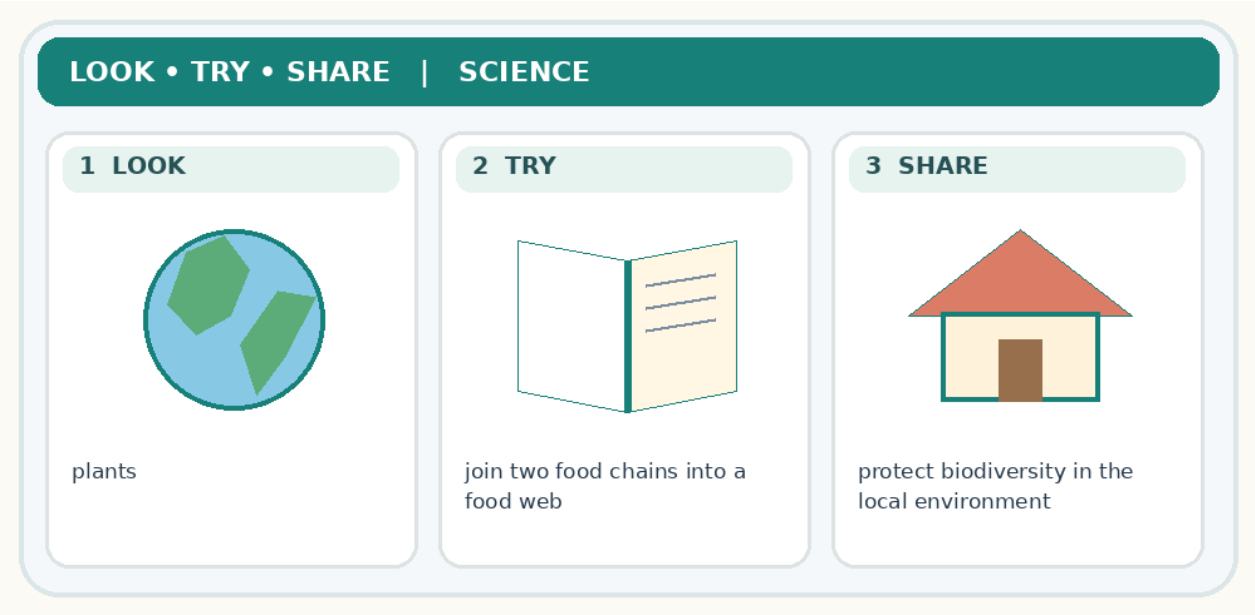
10. After practising Earth in the solar system, what is a useful next step?

A. leave the group without sharing B. share one observation and listen to feedback C. claim that feedback is never helpful D. tear up other learners work

Discuss the reasons for your choices with a partner. Practical performance and teacher observation matter as well as quiz answers.

Lesson 7. Food webs and balance

Systems | Term 2



EXPLORE

A food web contains several connected food chains.

Look for: plants, grasshopper, frog, snake

PLAY & APPLY

Activity: join two food chains into a food web

Materials: food-web cards and arrows

Real life: protect biodiversity in the local environment

Remember: Food webs are more complex than one chain.

Partner talk: What did you notice? Can you give a reason for your answer?

PLAY QUESTIONS

1. Which example is connected with this lesson about Food webs and balance?

A. pulley B. plants C. Moon D. petroleum

2. What is the best activity for practising Food webs and balance?

A. join two food chains into a food web B. ignore the instructions and rush C. leave all materials untouched D. copy without discussing anything

3. Which other example helps us explore Food webs and balance?

A. bones B. water filter C. grasshopper D. metal

4. Which materials should the group prepare to study Food webs and balance?

A. food-web cards and arrows B. nothing suitable for the task C. unsafe tools without an adult D. broken objects found at random

5. Where can you use what you learnt about Food webs and balance?

A. ignore the safety guidance B. protect biodiversity in the local environment C. damage the classroom display D. avoid talking about the idea

CHALLENGE QUESTIONS

1. Which explanation best describes our main idea in Food webs and balance?

A. A food web contains several connected food chains. B. Safety instructions can be ignored during an activity. C. An observation is exactly the same as a guess. D. One example proves that all cases are identical.

2. A classmate makes a mistake in Food webs and balance. Which reminder helps?

A. An observation is exactly the same as a guess. B. Food webs are more complex than one chain. C. One example proves that all cases are identical. D. Safety instructions can be ignored during an activity.

3. The class plans to explore Food webs and balance. Which action fits?

A. guess without looking at evidence B. join two food chains into a food web C. mix unrelated ideas without planning D. skip the task and claim it is complete

4. Choose a pair of examples that fits Food webs and balance.

A. plants and snake B. bandage and lungs C. Moon and pulley D. petroleum and coal

5. Which materials are most helpful for Food webs and balance?

A. sharp tools without supervision B. food-web cards and arrows C. unrelated broken electronics D. random objects that cannot help the task

6. How could a learner apply Food webs and balance outside school?

A. leave a mess after the lesson B. protect biodiversity in the local environment C. ignore school or home safety rules D. refuse to practise the lesson

7. Which key word should appear on a Food webs and balance revision card?

A. frog B. steel C. iron D. Earth

8. While studying Food webs and balance, which choice shows careful learning?

A. change the answer without reviewing B. observe, try, explain and check your result C. copy answers without understanding D. rush without listening

9. Which label belongs on a poster about Food webs and balance?

A. clean wound B. snake C. copper D. solar dryer

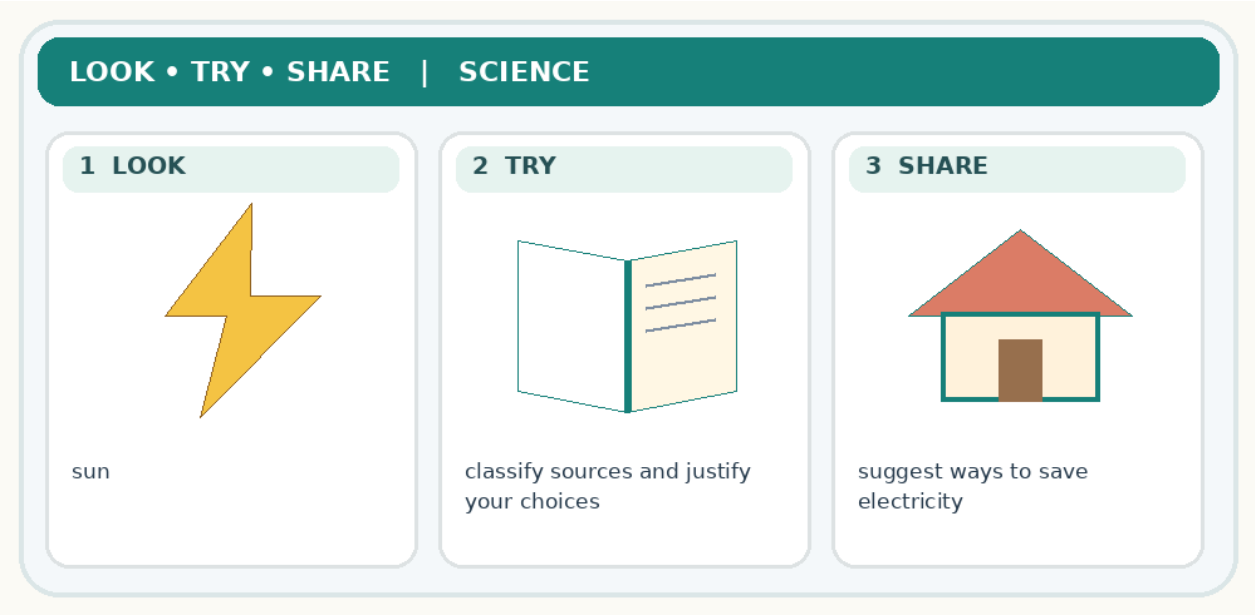
10. After practising Food webs and balance, what is a useful next step?

A. claim that feedback is never helpful B. tear up other learners work C. leave the group without sharing D. share one observation and listen to feedback

Discuss the reasons for your choices with a partner. Practical performance and teacher observation matter as well as quiz answers.

Lesson 8. Renewable and non-renewable energy

Forces and Energy | Term 2



EXPLORE

Some sources of energy replenish naturally; others are finite.

Look for: sun, wind, coal, petroleum

PLAY & APPLY

Activity: classify sources and justify your choices

Materials: energy picture cards

Real life: suggest ways to save electricity

Remember: Petroleum is not a renewable fuel.

Partner talk: What did you notice? Can you give a reason for your answer?

PLAY QUESTIONS

1. Which example is connected with this lesson about Renewable and non-renewable energy?

A. sun B. water filter C. food preservation D. heart

2. What is the best activity for practising Renewable and non-renewable energy?

A. classify sources and justify your choices B. ignore the instructions and rush C. leave all materials untouched D. copy without discussing anything

3. Which other example helps us explore Renewable and non-renewable energy?

A. clean wound B. muscles C. solar dryer D. wind

4. Which materials should the group prepare to study Renewable and non-renewable energy?

A. energy picture cards B. unsafe tools without an adult C. broken objects found at random D. nothing suitable for the task

5. Where can you use what you learnt about Renewable and non-renewable energy?

A. avoid talking about the idea B. damage the classroom display C. suggest ways to save electricity D. ignore the safety guidance

CHALLENGE QUESTIONS

1. Which explanation best describes our main idea in Renewable and non-renewable energy?

A. Some sources of energy replenish naturally; others are finite. B. One example proves that all cases are identical. C. Safety instructions can be ignored during an activity. D. An observation is exactly the same as a guess.

2. A classmate makes a mistake in Renewable and non-renewable energy. Which reminder helps?

A. Petroleum is not a renewable fuel. B. Safety instructions can be ignored during an activity. C. An observation is exactly the same as a guess. D. One example proves that all cases are identical.

3. The class plans to explore Renewable and non-renewable energy. Which action fits?

A. skip the task and claim it is complete B. mix unrelated ideas without planning C. guess without looking at evidence D. classify sources and justify your choices

4. Choose a pair of examples that fits Renewable and non-renewable energy.

A. sun and petroleum B. lungs and lever C. water filter and heart D. food preservation and clean water

5. Which materials are most helpful for Renewable and non-renewable energy?

A. energy picture cards B. random objects that cannot help the task C. unrelated broken electronics D. sharp tools without supervision

6. How could a learner apply Renewable and non-renewable energy outside school?

A. leave a mess after the lesson B. refuse to practise the lesson C. ignore school or home safety rules D. suggest ways to save electricity

7. Which key word should appear on a Renewable and non-renewable energy revision card?

A. seed B. coal C. vaccination D. plastic

8. While studying Renewable and non-renewable energy, which choice shows careful learning?

A. observe, try, explain and check your result B. change the answer without reviewing C. copy answers without understanding D. rush without listening

9. Which label belongs on a poster about Renewable and non-renewable energy?

A. handwashing B. Moon C. frog D. petroleum

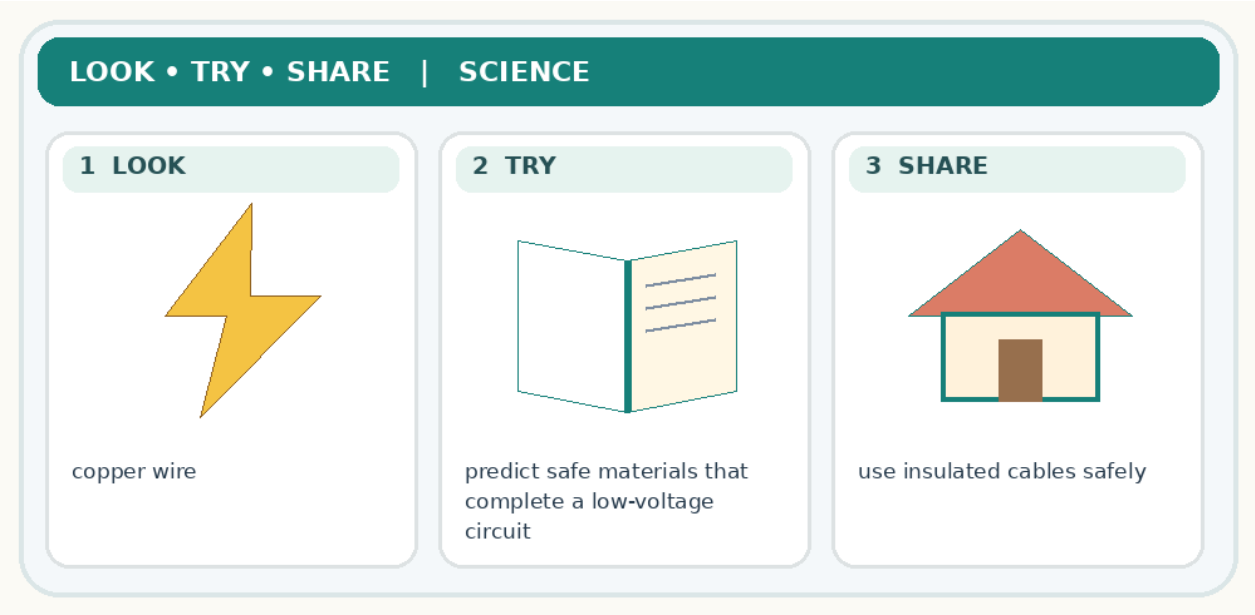
10. After practising Renewable and non-renewable energy, what is a useful next step?

A. claim that feedback is never helpful B. leave the group without sharing C. share one observation and listen to feedback D. tear up other learners work

Discuss the reasons for your choices with a partner. Practical performance and teacher observation matter as well as quiz answers.

Lesson 9. Conductors and circuits

Forces and Energy | Term 2



EXPLORE

Conductors allow current through and insulators resist its flow.

Look for: copper wire, rubber, plastic, metal

PLAY & APPLY

Activity: predict safe materials that complete a low-voltage circuit

Materials: battery circuit with adult supervision

Real life: use insulated cables safely

Remember: Never test materials using wall electricity.

Partner talk: What did you notice? Can you give a reason for your answer?

PLAY QUESTIONS

1. Which example is connected with this lesson about Conductors and circuits?

A. steel B. vaccination C. heart D. copper wire

2. What is the best activity for practising Conductors and circuits?

A. predict safe materials that complete a low-voltage circuit B. copy without discussing anything C. ignore the instructions and rush D. leave all materials untouched

3. Which other example helps us explore Conductors and circuits?

A. frog B. rubber C. clean wound D. rest

4. Which materials should the group prepare to study Conductors and circuits?

A. unsafe tools without an adult B. battery circuit with adult supervision C. nothing suitable for the task D. broken objects found at random

5. Where can you use what you learnt about Conductors and circuits?

A. use insulated cables safely B. avoid talking about the idea C. damage the classroom display D. ignore the safety guidance

CHALLENGE QUESTIONS

1. Which explanation best describes our main idea in Conductors and circuits?

A. Conductors allow current through and insulators resist its flow. B. Safety instructions can be ignored during an activity. C. One example proves that all cases are identical. D. An observation is exactly the same as a guess.

2. A classmate makes a mistake in Conductors and circuits. Which reminder helps?

A. Safety instructions can be ignored during an activity. B. One example proves that all cases are identical. C. An observation is exactly the same as a guess. D. Never test materials using wall electricity.

3. The class plans to explore Conductors and circuits. Which action fits?

A. skip the task and claim it is complete B. mix unrelated ideas without planning C. guess without looking at evidence D. predict safe materials that complete a low-voltage circuit

4. Choose a pair of examples that fits Conductors and circuits.

A. copper wire and metal B. steel and heart C. Mars and irrigation D. vaccination and sand

5. Which materials are most helpful for Conductors and circuits?

A. unrelated broken electronics B. sharp tools without supervision C. battery circuit with adult supervision D. random objects that cannot help the task

6. How could a learner apply Conductors and circuits outside school?

A. use insulated cables safely B. refuse to practise the lesson C. leave a mess after the lesson D. ignore school or home safety rules

7. Which key word should appear on a Conductors and circuits revision card?

A. shade B. grass root C. pollination D. plastic

8. While studying Conductors and circuits, which choice shows careful learning?

A. observe, try, explain and check your result B. copy answers without understanding C. change the answer without reviewing D. rush without listening

9. Which label belongs on a poster about Conductors and circuits?

A. clean drains B. clay C. metal D. adult help

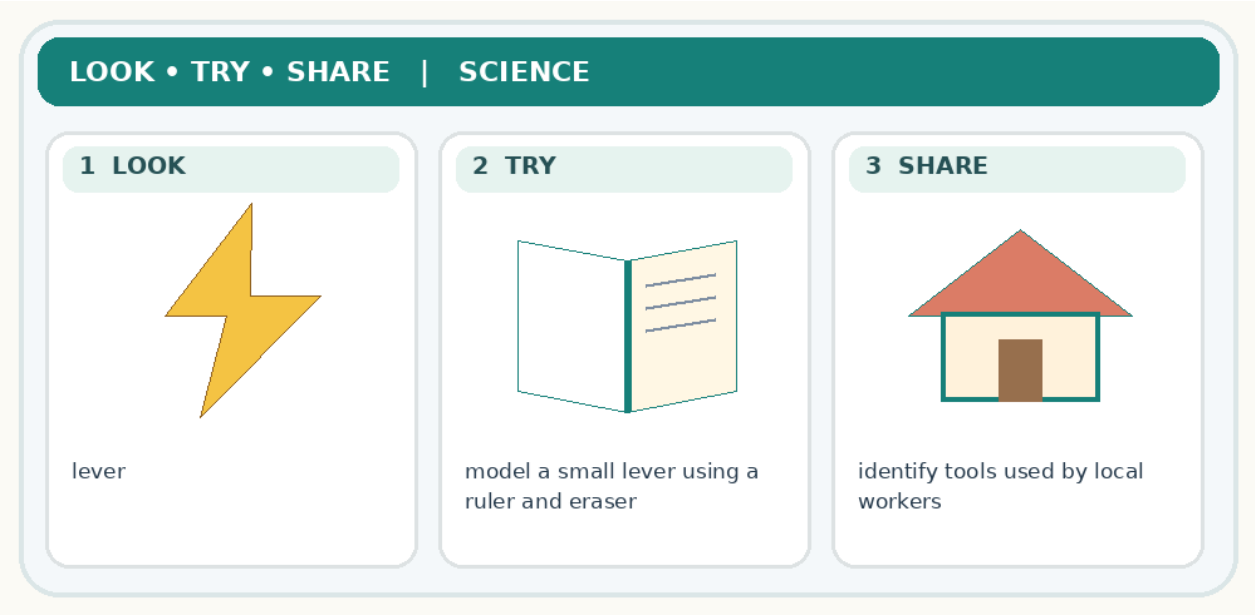
10. After practising Conductors and circuits, what is a useful next step?

A. claim that feedback is never helpful B. tear up other learners work C. leave the group without sharing D. share one observation and listen to feedback

Discuss the reasons for your choices with a partner. Practical performance and teacher observation matter as well as quiz answers.

Lesson 10. Levers and simple machines

Forces and Energy | Term 2



EXPLORE

Simple machines help us change the direction or size of a force.

Look for: lever, wheel, pulley, inclined plane

PLAY & APPLY

Activity: model a small lever using a ruler and eraser

Materials: ruler,eraser as fulcrum, picture cards

Real life: identify tools used by local workers

Remember: A lever does not create energy from nothing.

Partner talk: What did you notice? Can you give a reason for your answer?

PLAY QUESTIONS

1. Which example is connected with this lesson about Levers and simple machines?

A. clay B. handwashing C. Mars D. lever

2. What is the best activity for practising Levers and simple machines?

A. ignore the instructions and rush B. copy without discussing anything C. leave all materials untouched D. model a small lever using a ruler and eraser

3. Which other example helps us explore Levers and simple machines?

A. wheel B. vaccination C. solar dryer D. bean root

4. Which materials should the group prepare to study Levers and simple machines?

A. unsafe tools without an adult B. ruler,eraser as fulcrum, picture cards C. nothing suitable for the task D. broken objects found at random

5. Where can you use what you learnt about Levers and simple machines?

A. damage the classroom display B. avoid talking about the idea C. identify tools used by local workers D. ignore the safety guidance

CHALLENGE QUESTIONS

1. Which explanation best describes our main idea in Levers and simple machines?

A. An observation is exactly the same as a guess. B. Safety instructions can be ignored during an activity. C. Simple machines help us change the direction or size of a force. D. One example proves that all cases are identical.

2. A classmate makes a mistake in Levers and simple machines. Which reminder helps?

A. An observation is exactly the same as a guess. B. A lever does not create energy from nothing. C. One example proves that all cases are identical. D. Safety instructions can be ignored during an activity.

3. The class plans to explore Levers and simple machines. Which action fits?

A. guess without looking at evidence B. model a small lever using a ruler and eraser C. skip the task and claim it is complete D. mix unrelated ideas without planning

4. Choose a pair of examples that fits Levers and simple machines.

A. lever and inclined plane B. handwashing and plastic C. Mars and clay D. clean wound and snake

5. Which materials are most helpful for Levers and simple machines?

A. random objects that cannot help the task B. sharp tools without supervision C. ruler,eraser as fulcrum, picture cards D. unrelated broken electronics

6. How could a learner apply Levers and simple machines outside school?

A. identify tools used by local workers B. refuse to practise the lesson C. leave a mess after the lesson D. ignore school or home safety rules

7. Which key word should appear on a Levers and simple machines revision card?

A. aluminium B. pulley C. rest D. wind

8. While studying Levers and simple machines, which choice shows careful learning?

A. rush without listening B. change the answer without reviewing C. copy answers without understanding D. observe, try, explain and check your result

9. Which label belongs on a poster about Levers and simple machines?

A. inclined plane B. petroleum C. grasshopper D. Earth

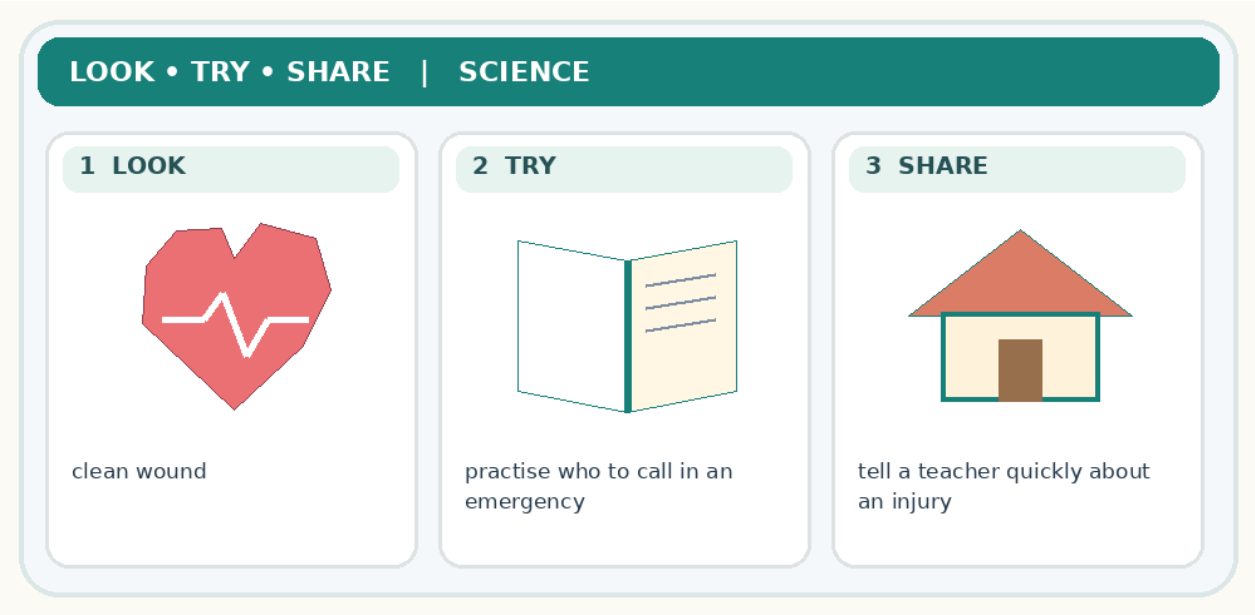
10. After practising Levers and simple machines, what is a useful next step?

A. share one observation and listen to feedback B. leave the group without sharing C. tear up other learners work D. claim that feedback is never helpful

Discuss the reasons for your choices with a partner. Practical performance and teacher observation matter as well as quiz answers.

Lesson 11. First aid and safety

Humans and the Environment | Term 3



EXPLORE

Basic first aid begins with getting adult help and keeping people safe.

Look for: clean wound, adult help, bandage, rest

PLAY & APPLY

Activity: practise who to call in an emergency

Materials: first aid picture cards

Real life: tell a teacher quickly about an injury

Remember: Never give medicine without proper adult guidance.

Partner talk: What did you notice? Can you give a reason for your answer?

PLAY QUESTIONS

1. Which example is connected with this lesson about First aid and safety?

A. muscles B. irrigation C. lever D. clean wound

2. What is the best activity for practising First aid and safety?

A. practise who to call in an emergency B. copy without discussing anything C. leave all materials untouched D. ignore the instructions and rush

3. Which other example helps us explore First aid and safety?

A. adult help B. heart C. Sun D. plants

4. Which materials should the group prepare to study First aid and safety?

A. nothing suitable for the task B. first aid picture cards C. broken objects found at random D. unsafe tools without an adult

5. Where can you use what you learnt about First aid and safety?

A. avoid talking about the idea B. damage the classroom display C. ignore the safety guidance D. tell a teacher quickly about an injury

CHALLENGE QUESTIONS

1. Which explanation best describes our main idea in First aid and safety?

A. Safety instructions can be ignored during an activity. B. An observation is exactly the same as a guess. C. One example proves that all cases are identical. D. Basic first aid begins with getting adult help and keeping people safe.

2. A classmate makes a mistake in First aid and safety. Which reminder helps?

A. One example proves that all cases are identical. B. Never give medicine without proper adult guidance. C. An observation is exactly the same as a guess. D. Safety instructions can be ignored during an activity.

3. The class plans to explore First aid and safety. Which action fits?

A. mix unrelated ideas without planning B. practise who to call in an emergency C. skip the task and claim it is complete D. guess without looking at evidence

4. Choose a pair of examples that fits First aid and safety.

A. clean wound and rest B. lungs and metal C. muscles and lever D. irrigation and iron

5. Which materials are most helpful for First aid and safety?

A. unrelated broken electronics B. first aid picture cards C. sharp tools without supervision D. random objects that cannot help the task

6. How could a learner apply First aid and safety outside school?

A. refuse to practise the lesson B. tell a teacher quickly about an injury C. ignore school or home safety rules D. leave a mess after the lesson

7. Which key word should appear on a First aid and safety revision card?

A. bandage B. shade C. Mars D. bones

8. While studying First aid and safety, which choice shows careful learning?

A. copy answers without understanding B. rush without listening C. observe, try, explain and check your result D. change the answer without reviewing

9. Which label belongs on a poster about First aid and safety?

A. rubber B. rest C. rain collection D. sun

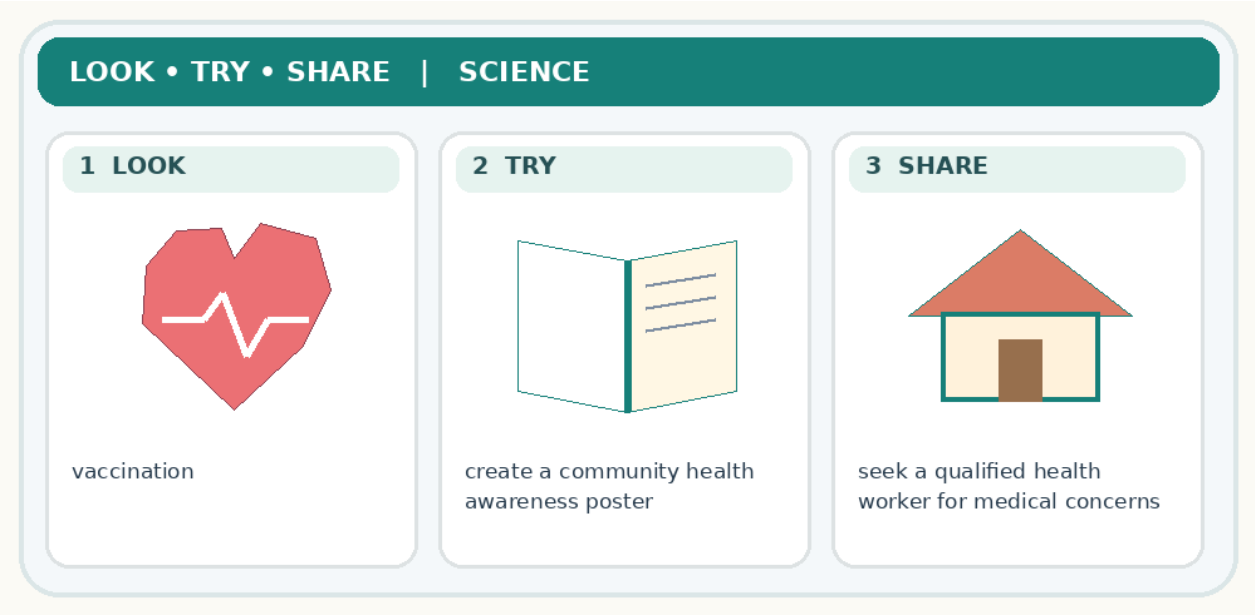
10. After practising First aid and safety, what is a useful next step?

A. claim that feedback is never helpful B. tear up other learners work C. share one observation and listen to feedback D. leave the group without sharing

Discuss the reasons for your choices with a partner. Practical performance and teacher observation matter as well as quiz answers.

Lesson 12. Infection prevention

Humans and the Environment | Term 3



EXPLORE

Vaccination, hygiene and safe water help protect communities.

Look for: vaccination, handwashing, clean water, mosquito net

PLAY & APPLY

Activity: create a community health awareness poster

Materials: health message cards

Real life: seek a qualified health worker for medical concerns

Remember: Vaccines are not a replacement for clean water and hygiene.

Partner talk: What did you notice? Can you give a reason for your answer?

PLAY QUESTIONS

1. Which example is connected with this lesson about Infection prevention?

A. carrot root B. wind C. iron D. vaccination

2. What is the best activity for practising Infection prevention?

A. create a community health awareness poster B. ignore the instructions and rush C. copy without discussing anything D. leave all materials untouched

3. Which other example helps us explore Infection prevention?

A. handwashing B. sand C. Sun D. irrigation

4. Which materials should the group prepare to study Infection prevention?

A. broken objects found at random B. unsafe tools without an adult C. health message cards D. nothing suitable for the task

5. Where can you use what you learnt about Infection prevention?

A. seek a qualified health worker for medical concerns B. avoid talking about the idea C. ignore the safety guidance D. damage the classroom display

CHALLENGE QUESTIONS

1. Which explanation best describes our main idea in Infection prevention?

A. Vaccination, hygiene and safe water help protect communities. B. Safety instructions can be ignored during an activity. C. One example proves that all cases are identical. D. An observation is exactly the same as a guess.

2. A classmate makes a mistake in Infection prevention. Which reminder helps?

A. An observation is exactly the same as a guess. B. Safety instructions can be ignored during an activity. C. One example proves that all cases are identical. D. Vaccines are not a replacement for clean water and hygiene.

3. The class plans to explore Infection prevention. Which action fits?

A. guess without looking at evidence B. mix unrelated ideas without planning C. skip the task and claim it is complete D. create a community health awareness poster

4. Choose a pair of examples that fits Infection prevention.

A. vaccination and mosquito net B. clean drains and water filter C. iron and Earth D. wind and carrot root

5. Which materials are most helpful for Infection prevention?

A. unrelated broken electronics B. sharp tools without supervision C. health message cards D. random objects that cannot help the task

6. How could a learner apply Infection prevention outside school?

A. leave a mess after the lesson B. refuse to practise the lesson C. seek a qualified health worker for medical concerns D. ignore school or home safety rules

7. Which key word should appear on a Infection prevention revision card?

A. clean water B. petroleum C. steel D. rest

8. While studying Infection prevention, which choice shows careful learning?

A. change the answer without reviewing B. observe, try, explain and check your result C. rush without listening D. copy answers without understanding

9. Which label belongs on a poster about Infection prevention?

A. clean wound B. mosquito net C. metal D. adult help

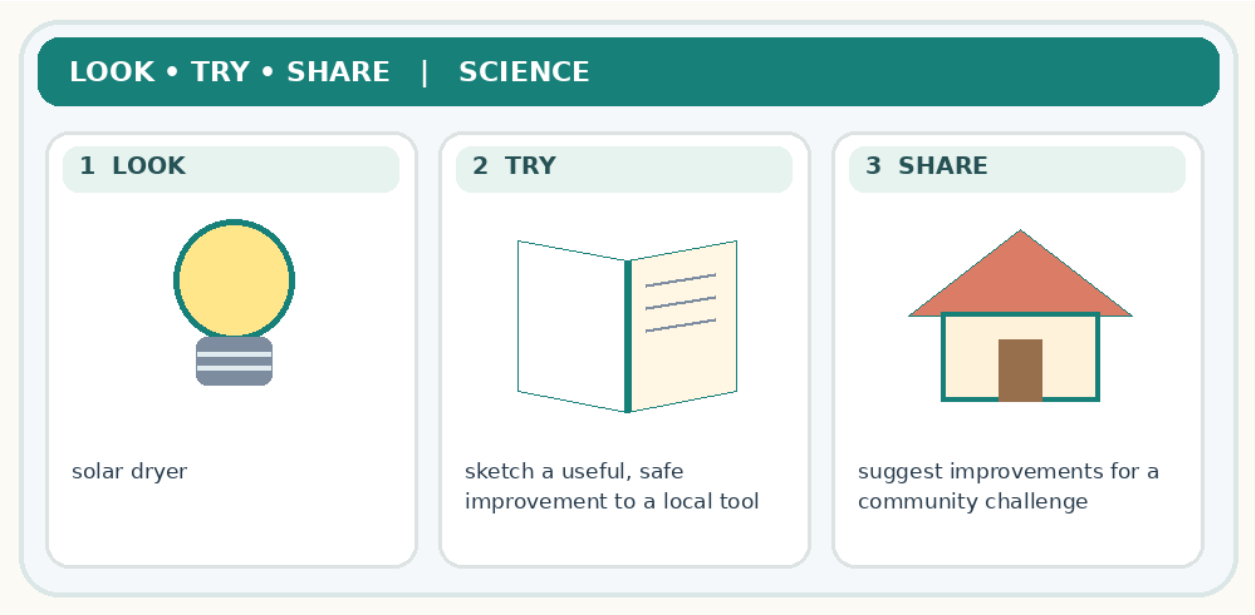
10. After practising Infection prevention, what is a useful next step?

A. share one observation and listen to feedback B. claim that feedback is never helpful C. tear up other learners work D. leave the group without sharing

Discuss the reasons for your choices with a partner. Practical performance and teacher observation matter as well as quiz answers.

Lesson 13. Innovation in Ghana

Humans and the Environment | Term 3



EXPLORE

Research and practical design improve work and daily life.

Look for: solar dryer, irrigation, water filter, food preservation

PLAY & APPLY

Activity: sketch a useful, safe improvement to a local tool

Materials: recycled craft materials

Real life: suggest improvements for a community challenge

Remember: Innovation begins with identifying a problem.

Partner talk: What did you notice? Can you give a reason for your answer?

PLAY QUESTIONS

1. Which example is connected with this lesson about Innovation in Ghana?

A. rain collection B. muscles C. sand D. solar dryer

2. What is the best activity for practising Innovation in Ghana?

A. leave all materials untouched B. copy without discussing anything C. ignore the instructions and rush D. sketch a useful, safe improvement to a local tool

3. Which other example helps us explore Innovation in Ghana?

A. irrigation B. aluminium C. loam D. plants

4. Which materials should the group prepare to study Innovation in Ghana?

A. nothing suitable for the task B. recycled craft materials C. broken objects found at random D. unsafe tools without an adult

5. Where can you use what you learnt about Innovation in Ghana?

A. ignore the safety guidance B. avoid talking about the idea C. suggest improvements for a community challenge D. damage the classroom display

CHALLENGE QUESTIONS

1. Which explanation best describes our main idea in Innovation in Ghana?

A. An observation is exactly the same as a guess. B. One example proves that all cases are identical. C. Safety instructions can be ignored during an activity. D. Research and practical design improve work and daily life.

2. A classmate makes a mistake in Innovation in Ghana. Which reminder helps?

A. Innovation begins with identifying a problem. B. One example proves that all cases are identical. C. Safety instructions can be ignored during an activity. D. An observation is exactly the same as a guess.

3. The class plans to explore Innovation in Ghana. Which action fits?

A. sketch a useful, safe improvement to a local tool B. guess without looking at evidence C. mix unrelated ideas without planning D. skip the task and claim it is complete

4. Choose a pair of examples that fits Innovation in Ghana.

A. sand and heart B. muscles and rain collection C. solar dryer and food preservation D. rubber and grass root

5. Which materials are most helpful for Innovation in Ghana?

A. recycled craft materials B. random objects that cannot help the task C. sharp tools without supervision D. unrelated broken electronics

6. How could a learner apply Innovation in Ghana outside school?

A. refuse to practise the lesson B. ignore school or home safety rules C. leave a mess after the lesson D. suggest improvements for a community challenge

7. Which key word should appear on a Innovation in Ghana revision card?

A. adult help B. water filter C. Sun D. maize root

8. While studying Innovation in Ghana, which choice shows careful learning?

A. rush without listening B. observe, try, explain and check your result C. change the answer without reviewing D. copy answers without understanding

9. Which label belongs on a poster about Innovation in Ghana?

A. steel B. food preservation C. metal D. seed

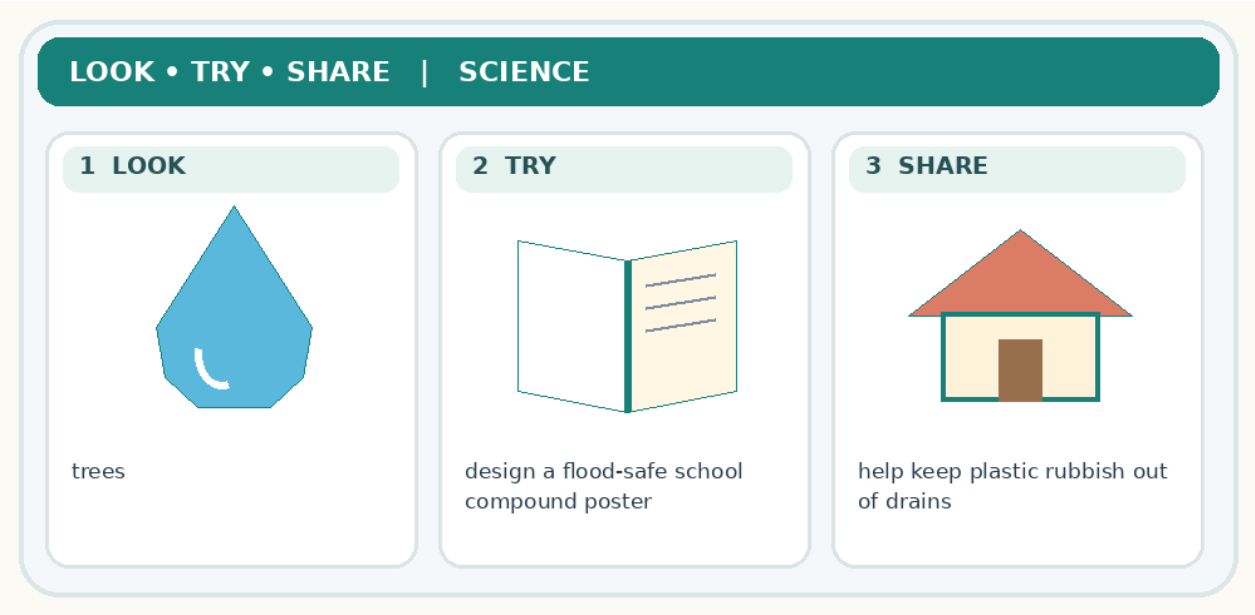
10. After practising Innovation in Ghana, what is a useful next step?

A. share one observation and listen to feedback B. leave the group without sharing C. claim that feedback is never helpful D. tear up other learners work

Discuss the reasons for your choices with a partner. Practical performance and teacher observation matter as well as quiz answers.

Lesson 14. Climate resilience

Humans and the Environment | Term 3



EXPLORE

Communities can prepare for heat, floods and changing weather patterns.

Look for: trees, clean drains, shade, rain collection

PLAY & APPLY

Activity: design a flood-safe school compound poster

Materials: community map and pencils

Real life: help keep plastic rubbish out of drains

Remember: Cleaning drains does not eliminate all flood risk.

Partner talk: What did you notice? Can you give a reason for your answer?

PLAY QUESTIONS

1. Which example is connected with this lesson about Climate resilience?

A. trees B. petroleum C. copper wire D. lungs

2. What is the best activity for practising Climate resilience?

A. leave all materials untouched B. ignore the instructions and rush C. copy without discussing anything D. design a flood-safe school compound poster

3. Which other example helps us explore Climate resilience?

A. clean drains B. Earth C. flower D. copper

4. Which materials should the group prepare to study Climate resilience?

A. community map and pencils B. nothing suitable for the task C. broken objects found at random D. unsafe tools without an adult

5. Where can you use what you learnt about Climate resilience?

A. help keep plastic rubbish out of drains B. ignore the safety guidance C. avoid talking about the idea D. damage the classroom display

CHALLENGE QUESTIONS

1. Which explanation best describes our main idea in Climate resilience?

A. One example proves that all cases are identical. B. Safety instructions can be ignored during an activity. C. An observation is exactly the same as a guess. D. Communities can prepare for heat, floods and changing weather patterns.

2. A classmate makes a mistake in Climate resilience. Which reminder helps?

A. One example proves that all cases are identical. B. Safety instructions can be ignored during an activity. C. Cleaning drains does not eliminate all flood risk. D. An observation is exactly the same as a guess.

3. The class plans to explore Climate resilience. Which action fits?

A. mix unrelated ideas without planning B. design a flood-safe school compound poster C. skip the task and claim it is complete D. guess without looking at evidence

4. Choose a pair of examples that fits Climate resilience.

A. trees and rain collection B. lungs and petroleum C. copper wire and grasshopper D. rest and bones

5. Which materials are most helpful for Climate resilience?

A. sharp tools without supervision B. random objects that cannot help the task C. unrelated broken electronics D. community map and pencils

6. How could a learner apply Climate resilience outside school?

A. leave a mess after the lesson B. ignore school or home safety rules C. refuse to practise the lesson D. help keep plastic rubbish out of drains

7. Which key word should appear on a Climate resilience revision card?

A. Sun B. shade C. handwashing D. metal

8. While studying Climate resilience, which choice shows careful learning?

A. observe, try, explain and check your result B. change the answer without reviewing C. copy answers without understanding D. rush without listening

9. Which label belongs on a poster about Climate resilience?

A. plastic B. plants C. rain collection D. solar dryer

10. After practising Climate resilience, what is a useful next step?

A. claim that feedback is never helpful B. tear up other learners work C. leave the group without sharing D. share one observation and listen to feedback

Discuss the reasons for your choices with a partner. Practical performance and teacher observation matter as well as quiz answers.

TEACHER ANSWER KEY

Lesson 1 — Tap roots and fibrous roots: 1:B 2:B 3:B 4:A 5:B 6:B 7:D 8:A 9:B 10:C 11:D 12:C 13:A 14:A 15:D

Lesson 2 — Properties of metals: 1:C 2:B 3:C 4:B 5:A 6:A 7:C 8:C 9:B 10:A 11:A 12:D 13:A 14:B 15:A

Lesson 3 — Soil and rocks: 1:D 2:C 3:C 4:D 5:B 6:A 7:B 8:A 9:D 10:C 11:C 12:C 13:C 14:A 15:B

Lesson 4 — Plant life cycles and reproduction: 1:D 2:B 3:A 4:C 5:B 6:D 7:C 8:D 9:C 10:C 11:A 12:C 13:C 14:A 15:A

Lesson 5 — Body systems working together: 1:B 2:B 3:B 4:A 5:D 6:A 7:C 8:B 9:A 10:A 11:D 12:A 13:D 14:A 15:D

Lesson 6 — Earth in the solar system: 1:D 2:B 3:D 4:D 5:A 6:D 7:B 8:C 9:D 10:C 11:B 12:D 13:D 14:A 15:B

Lesson 7 — Food webs and balance: 1:B 2:A 3:C 4:A 5:B 6:A 7:B 8:B 9:A 10:B 11:B 12:A 13:B 14:B 15:D

Lesson 8 — Renewable and non-renewable energy: 1:A 2:A 3:D 4:A 5:C 6:A 7:A 8:D 9:A 10:A 11:D 12:B 13:A 14:D 15:C

Lesson 9 — Conductors and circuits: 1:D 2:A 3:B 4:B 5:A 6:A 7:D 8:D 9:A 10:C 11:A 12:D 13:A 14:C 15:D

Lesson 10 — Levers and simple machines: 1:D 2:D 3:A 4:B 5:C 6:C 7:B 8:B 9:A 10:C 11:A 12:B 13:D 14:A 15:A

Lesson 11 — First aid and safety: 1:D 2:A 3:A 4:B 5:D 6:D 7:B 8:B 9:A 10:B 11:B 12:A 13:C 14:B 15:C

Lesson 12 — Infection prevention: 1:D 2:A 3:A 4:C 5:A 6:A 7:D 8:D 9:A 10:C 11:C 12:A 13:B 14:B 15:A

Lesson 13 — Innovation in Ghana: 1:D 2:D 3:A 4:B 5:C 6:D 7:A 8:A 9:C 10:A 11:D 12:B 13:B 14:B 15:A

Lesson 14 — Climate resilience: 1:A 2:D 3:A 4:A 5:A 6:D 7:C 8:B 9:A 10:D 11:D 12:B 13:A 14:C 15:D